

Approach to a child with cervical lymphadenopathy

Aim and scope of this document:

This document Aim to help pediatrician and general practitioner to deal with and investigate a child with cervical lymphadenopathy?

What is lymphadenopathy?

The presence of one or more lymph nodes of more than 1 cm in diameter, with or without an abnormality in character

History

The following questions are important in evaluating for the various possible causes of neck masses in children:

- Age of the patient
- Duration of enlarged neck mass: (a) present since birth or recently discovered, and for how long (b) is the neck mass recurrent or not?
- Is there any associated symptoms: (a) focal symptoms (e.g. pain, erythema); (b) infective symptoms (e.g., fever, sore throat, other localizing symptoms of infection); (c) rashes, arthralgia; and (d) weight loss, loss of appetite, night sweats, fatigue, easy bruising and pallor
- Recent illnesses (e.g. pharyngitis, tonsillitis, dental infections)
- History of recurrent infections, recurrent skin abscesses or recurrent lymphadenopathy
- medications history, including any treatment with antibiotics
- Exposure to animals (particularly cats), insects, sick persons, persons with tuberculosis or chronic cough
 - Exposure to sick person
- Recent travel history
- Immunization status

Differential diagnoses of neck masses:

Location	Congenital	Acquired
Anterior midline or paramedian	• Thyroglossal cyst • Dermoid cyst • Thymic cyst	• _ Thyroid lesions: goitre, thyroiditis, rarely thyroid cancer
Anterior Triangle	• _ Branchial cleft anomalies • _ Congenital muscular torticollis • _ Vascular lesions (haemangioma, lymphangioma) • Cervical rib	• Lymphadenopathy • _ Submandibular gland enlargement (e.g. sialadenitis, sialolithiasis, rarely neoplasm) • Epidermoid cyst
Angle of the mandible		• _ Parotid gland enlargement (e.g. acute or recurrent parotitis, sialolithiasis, rarely neoplasm)
Posterior Triangle	Cystic hygroma	• Lymphadenopathy • Epidermoid cyst • _ Malignant neoplasm (e.g. rhabdomyosarcoma, neuroblastoma)

Physical examination:

- Check general health and growth of the child with inspection of the child's growth parameters in growth charts
- General examination of pallor, cubbing or jaundice
- Local examination of the neck: location, size, shape, fluctuance, consistency, skin changes overlaying the lymph nodes, and tenderness, warmth overlying skin, mobility of the lymph nodes
- A sequential examination of other cervical lymph nodes
- Examination of other areas of peripheral lymphadenopathy should also be palpated (i.e. axillary, epitrochlear, inguinal, popliteal) to determine if the lymphadenopathy is localized or generalized.

Presence of Supraclavicular lymphadenopathy is concerning, as it may be associated with an underlying malignancy

- Examination of eyes, ears, nose, mouth and throat for possible source of infection that could be cause of the cervical lymphadenopathy
- Examination for any joint or bony pain, or presence of skin rash and abdominal examination for any organomegaly

Acute cervical lymphadenopathy

Infective causes

Acute lymphadenopathy is a self-limiting, transient response of lymphatic tissue hyperplasia to a local infective process that improves with resolution of the viral illness.

Causes of acute cervical lymphadenopathy

Duration	Causes	Associated clinical features
Acute (< 2 weeks)	Infections (more common) • Viral infections – respiratory viruses (adenovirus, influenza virus and respiratory syncytial virus etc)	Bilateral cervical lymphadenopathy, upper respiratory tract infection symptoms
	• EBV and CMV	- More common in older children of school-going age - EBV – fever, lethargy, sore throat, exudative tonsillitis, generalized lymphadenopathy Hepatosplenomegaly
	Bacterial infections – Staphylococcus aureus, Group A Streptococcus	- Often affects children aged 1–4 years -Unilateral, tender lymphadenopathy, overlying erythema, fever
	Group B Streptococcus	Neonatal/infant age group
	Anaerobic bacteria	Oral mucositis, dental caries or infections
	Immunologic Kawasaki disease	- Affects children under five years of age. -Fever, rash, red lips, strawberry tongue, unilateral cervical lymphadenopathy, swollen or erythematous hands or feet, BCG vaccination scar erythema or induration
	Kikuchi-Fujimoto disease (histiocytic necrotising Lymphadenitis)	- Self-limiting, benign condition of unknown etiology - Most commonly aged 7–12 years

		-Prolonged, unexplained fever, chills, weight loss, night sweats, rash and Arthralgia - Complications including aseptic meningitis, -Reports of progress to systemic lupus erythematosus - Biopsy for diagnosis
	Others: Adverse drug reaction • Histiocytosis	- Phenytoin, isoniazid, pyrimethamine and allopurinol.

- If the patient with acute cervical lymphadenopathy does not improve after 4–7 days, then think of primary bacterial or secondary bacterial lymphadenitis

Causes of subacute cervical lymphadenopathy

Duration	Causes	Associated clinical features
Subacute to chronic (2–6 weeks)	<u>Infections (more common)</u> Non-tuberculous mycobacterium	-Unilateral, usually multiple, firm lymph nodes; thin and violaceous overlying skin. May develop fluctuance or cutaneous sinus tracts - Often affects children aged 1–5 years - It commonly affects the upper anterior cervical and submandibular lymph nodes
	EBV, CMV	EBV – fever, lethargy, sore throat, exudative tonsillitis, Hepatosplenomegaly
	Mycobacterium tuberculosis	-Unilateral or bilateral, firm, non-tender lymph nodes; may be fixed to underlying tissues and progress to become matted, low-grade fever with systemic symptoms - Preceding history of exposure to tuberculosis - 28%–71% of affected patients have abnormal chest radiographs

	Bartonella henselae (cat-scratch disease)	- Cat scratch or bite - causes a granulomatous infection
	HIV	- look for other HIV related features
	Fungal or protozoan infection (e.g. toxoplasmosis, malaria)	
	<u>Immunologic (less common)</u> Systemic lupus erythematosus	- Generalized non-tender lymphadenopathy, fever, rash, arthralgia, arthritis, hepatosplenomegaly - Lymph nodes are often non-tender and discrete, with a range of sizes from very small to a few centimeters in diameter
	Juvenile idiopathic arthritis	
	Dermatomyositis	

Causes of chronic cervical lymphadenopathy

Duration	Causes	Associated clinical features
Chronic (> 6 weeks)	Malignancies (more common) Hodgkin's and non-Hodgkin's lymphoma	Firm, indurated, fixed, matted, usually non-tender lymph nodes; supraclavicular lymph nodes; systemic symptoms; petechiae; pallor; hepatosplenomegaly
	Acute leukaemia	
	Rhabdomyosarcoma	
	Neuroblastoma	
	Immunologic (less common) Common variable immunodeficiency	
	Chronic granulomatous disease	
	Hyper-immunoglobulin E syndrome	
	Miscellaneous (less common) Sarcoidosis	
	Storage disease (Gaucher disease, Niemann-Pick disease)	

Other conditions mimic cervical lymphadenopathy		
Salivary glands: sialadenitis	Most common organism being S aureus	- think of salivary gland duct obstruction -May need ENT referral
Adenoid and tonsillar enlargement	-Associated with recurrent throat infection - may have obstructive symptoms	-need ENT referral
Other differential diagnosis of cervical lymphadenopathy : • Mumps • Thyroglossal cyst • Branchial cleft cyst • Sternomastoid tumor • Cervical rib • Cystic hygroma • Hemangioma • Laryngocele • Dermoid cyst Data from Leung & Robson (1991)		

Red flags associated with cervical lymphadenopathy:

• Lymphadenopathy > 2 cm • Steady increase in size over 2-3 weeks • No improvement or decrease in size after 4-6 weeks • Supraclavicular lymphadenopathy • Hard, fixed, matted, non-tender lymphadenopathy • Persistent fever lasting more than one week • Signs and symptoms suggestive of malignancy: weight loss, petechiae, pallor, night sweats, hepatosplenomegaly • Signs and symptoms suggestive of autoimmune disease: rash, arthralgia, arthritis, generalized lymphadenopathy • Features suggestive of Kawasaki disease • Abnormalities in full blood count or chest radiography

INVESTIGATIONS

Blood investigations	
Full blood count	- Raised total white cell count and neutrophilia are suggestive of a bacterial etiology - cytopenia can be seen in viral aetiologies, hematological malignancies or autoimmune Conditions
Peripheral blood film	

	- Thrombocytosis might be seen in cases of reactive cervical lymphadenitis or inflammatory conditions such as KD - lymphopenia can be seen in systemic lupus erythematosus - Atypical lymphocytes or monocytes can be seen in infectious mononucleosis - Presence of blast cells in cases of hematological malignancies
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Blood investigations	
Erythrocyte sedimentation rate (ESR)	- Raised in infective (more likely from a bacterial cause) or inflammatory causes of cervical lymphadenopathy - Markedly raised ESR in relation to the CRP level would be more concerning for inflammatory conditions
C-reactive protein (CRP)	

Blood investigations	
Uric acid and lactate dehydrogenase (LDH)	- Raised uric acid and LDH levels are indicative of a high cell turnover state - Significantly raised levels are worrisome for an underlying malignancy
Liver function test (LFT) and renal function test	- Useful for determining multi-systemic involvement from the underlying condition
Other blood tests	- should be guided by history, physical examinations and clinical suspicion

Blood investigations	
- Serologic tests: EBV, CMV, Bartonella henselae and toxoplasmosis - Anti-streptolysin O titer test can be used in the diagnosis of a streptococcal infection	

Images	
Chest radiography	- To assess for evidence of mediastinal widening for cases of chronic lymphadenopathy - Mediastinal widening may be suggestive of a mediastinal mass, or hilar adenopathy
Ultrasonography	- First line modality to evaluate cervical lymphadenopathy - useful in determining if complications such as an abscess formation have occurred - useful to differentiate lymphadenopathy

	from other causes of congenital or acquired neck masses -Can also be used to guide surgical drainage in such cases
Computed tomography (CT) or magnetic resonance (MR)	- Useful in evaluating neck masses located within the deep neck

<i>Diagnostic biopsy: excisional biopsy</i>	
Indications: (a) persistent or progressive lymphadenopathy after 4–6 weeks; (b) lymph nodes greater than 2 cm; (c) symptoms and signs suggestive of malignancy; (d) supraclavicular lymph node; (e) unexplained, prolonged fever; (f) abnormal blood film concerning of malignancy (in particular cell line defects) (g) Abnormal chest radiography of malignancy (h) concerning features on ultrasonography or CT.	
Biopsy specimen can be sent for <u>histology, gram stain and bacterial culture, acid-fast bacilli (AFB) smear and culture, AFB polymerase chain reaction test, fungal smear and fungal culture.</u>	

<u>Management</u>	
Reactive cervical lymphadenopathy presents with isolated bilateral, mobile, small sub-centimeters lymphadenopathy in response to a self-limiting viral illness	Explain diagnosis and reassurance
Acute non-toxic cervical lymphadenitis	Oral antibiotic with follow up at 72 hours
Isolated cervical lymphadenopathy of 1–2 cm with no other red flag features	Follow up for 4–6 weeks until the swelling
Large cervical lymph nodes and/or red flags	Need urgent evaluation for biopsy

Reference:

- 1- Chang SSY, Xiong M, How CH, Lee DM. An approach to cervical lymphadenopathy in children. Singapore Med J. 2020 Nov;61(11):569-577. doi: 10.11622/smedj.2020151. PMID: 33283242; PMCID: PMC8040913.
- 2- Alexander K.C Leung, W.Lane M Robson, Childhood cervical lymphadenopathy, Journal of Pediatric Health Care, Volume 18, Issue 1, 2004, Pages 3-7, ISSN 0891-5245, <https://doi.org/10.1016/j.pedhc.2003.08.008>